25901609 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCBIF100 - BIOMOLECULES AND MOLECULAR VISUALIZATION

(2024 ADMISSION ONWARDS)

Duration: 1 Hour **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Listout the four levels of protein structure. [K] / [CO1]
2. What is a nucleoside? [K] / [CO2]
3. Which force is responsible for base pairing in DNA? [K] / [CO2]
4. Report two key applications of bioinformatics. [A] / [CO3]
5. Focus two main uses of BLAST in genomics. [An] / [CO4]
6. Determine basic features of Webmol. [A] / [CO5, CO6]
7. Determine major application of ORF Finder. [A] / [CO5, CO6]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **3** questions

Each question carries **5** marks

8. Listout main types of lipids found in human body with it's function? [U] / [CO1]

9. Differentiate A DNA and Z DNA? [U] / [CO2]

10. Explain the importance of MSA in identifying conserved protein domains. [An] / [CO4]

11. Explain how the Surface model is useful in visualizing molecular structures. [A] / [CO5, CO6]

12. Determine key features and functionalities of LocusLinker in bioinformatics [A] / [CO5, CO6]

[5x3 = 15]

Part C

Long essay Questions

Answer any **1** questions

Each question carries **10** marks

13. Differentiate monosaccharides and disaccharides? Give examples? [A] / [CO1]

14. Choose a database and explain its significance in genomic research. [A] / [CO3]

15. Write about molecular graphics and its types. [A] / [CO5, CO6]

[10x1 = 10]